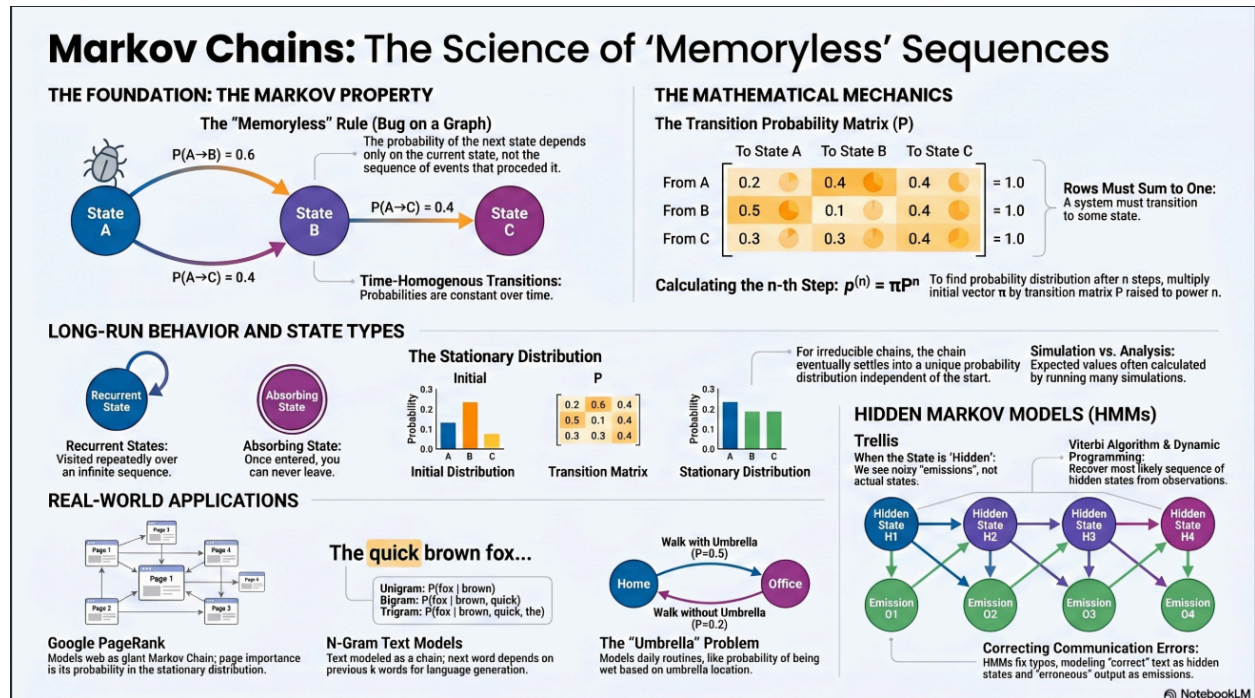


Kim, Andrew Min Yeop P.
LLM: Google NotebookLM

Original Prompt

Create an infographic on Markov Chains. It must be detailed yet easy to digest. Include visuals to supplement the lessons for a more complete understanding of the topic.



AI Enhancement Prompt to ChatGPT 5.3 Instant:

You are a master prompt engineer. Improve my prompt to create a superior output. This here is my prompt to noteboooklm in infographic generation. Create an infographic on Markov Chains. It must be detailed yet easy to digest. Include visuals to supplement the lessons for a more complete understanding of the topic.

AI Enhanced Prompt

Create a **high-quality infographic** explaining **Markov Chains** for students with basic probability knowledge.

Goals:

- Make it **visually intuitive, logically structured**, and **easy to follow**
- Balance **conceptual clarity** with **technical depth**

Content Structure (strictly follow this order):

- 1. Title Section**
 - Clear title: *"Markov Chains Explained"*
 - One-sentence definition (simple and memorable)
- 2. Core Idea**
 - Explain the **Markov Property** (memoryless concept)
 - Include a short real-life analogy (e.g., weather, board games)
- 3. Visual Diagram**
 - A **state diagram** with:
 - At least 3 states
 - Arrows labeled with transition probabilities
 - Ensure probabilities from each state sum to 1
- 4. Transition Matrix**
 - Show the matrix form of the example
 - Brief explanation of how to read it
- 5. Step-by-Step Example**
 - Demonstrate how to move from one state to another
 - Show at least one multi-step transition
- 6. Key Concepts Box**
 - States
 - Transition probabilities
 - Matrix representation
 - Steady-state (brief intro only)
- 7. Why It Matters**
 - 3–4 real-world applications (e.g., search engines, text prediction, finance)
- 8. Design Instructions**
 - Use clean layout with sections
 - Include icons or simple illustrations for each concept
 - Use color coding for states and transitions
 - Avoid dense paragraphs—use bullets and labels

Tone & Style:

- Clear, concise, and student-friendly
- Avoid unnecessary jargon, but keep terms accurate
- Each section should be understandable in under 20 seconds

Output Format:

- Structured infographic content with labeled sections
- Include descriptions of visuals (so they can be rendered)

Markov Chains: The Memoryless Path

A Markov Chains Explained
A mathematical system that hops between different "states" where the next step depends entirely on your current position, not your past path.



The Memoryless Property
 $P(X_{t+1} | X_t, X_{t-1}, \dots, X_0) = P(X_{t+1} | X_t)$
The past is irrelevant, only the present matters for predicting the future.



The Board Game Analogy
Your next position depends only on your current square and the roll of the die, not whether you climbed a ladder three turns ago.



Key Concepts

States & Probabilities

States are the possible conditions; **Transition Probabilities** are the chances of moving between them.

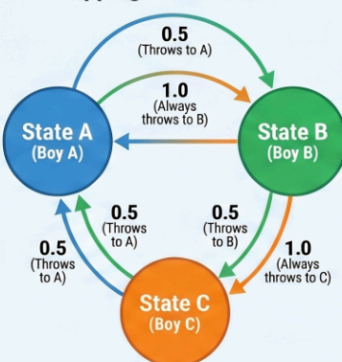
Absorbing vs. Recurrent

An **Absorbing State** is a "trap" you can't leave (like winning a game); a **Recurrent State** is one the system will eventually visit again.

Steady-State (Stationary Distribution)

The long-run probability distribution where the chances of being in each state stop changing, regardless of where you started.

Mapping the Transitions



The "Sum to One" Rule

For any single state, the sum of all outgoing probabilities must equal 1 (or 100%), because the system must go somewhere.

The Transition Matrix

The Matrix Form (P)

A "Stochastic Matrix" where the entry in row i and column j represents the probability of moving from state i to state j .

Reading the Grid

Rows represent your "Current State" and columns represent your "Next State." Row 1 of the Ball Throwing example would be $[0, 1, 0]$.

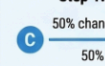
Current State	Next State		
	A	B	C
A	0	1	0
B	0	0	1
C	0.5	0.5	0

Moving Two Steps Ahead

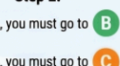
To find the probability of being in a state after n steps, you calculate the n -th power of the matrix (P^n).

Formally: $P(X_{t+1} | X_t, X_{t-1}, \dots, X_0) = P(X_{t+1} | X_t)$

Step 1:



Step 2:



After 2 steps, you have a 50% chance of being at B and 50% at C.

Why It Matters



Google PageRank
Ranks web pages by simulating a random walk (a Markov Chain) across the internet to see which pages are visited most often.



Text Prediction
"n-gram" models predict the next letter or word in a sentence based on the frequency of previous characters in a dataset.



Prediction
Finants task predict the letter or monil a sentences predict an identairilly and althrough a maze.



Finance & Science
Used to model stock market fluctuations, genetics (offspring traits), and even the path of a mouse through a maze.

NotebookLM

Reflection:

At the start of this lab, I thought about building a complex setup—like connecting an AI server to a business tool like Qlik. While I could have done it, I realized it would make things unnecessarily complicated. It risked making the project more about setting up systems rather than actually learning the content. Because of this, I switched to a more straightforward and practical idea: using Google NotebookLM to create a study-guide infographic.

Looking at the outputs before and after fixing my prompt made the power of prompt engineering clear. The original prompt gave an infographic that was mostly correct, but it lacked structure and a central visual theme. The updated prompt created a much more organized, visual, and helpful study tool.

This experience taught me that AI tools don't just magically give perfect answers on their own. The quality of what they produce depends entirely on how clear and exact their instructions are. By explicitly guiding the AI's reasoning, layout, and level of detail, I was able to turn dense text from four books into a condensed, useful learning tool.

Overall, this activity reframed how I approach AI. Instead of seeing these tools as machines that give instant answers in one step, I now see them as systems that need well-designed directions to work their best. I learned that prompt engineering is about setting clear rules that guide the AI toward the exact output one needs.